Cloud Computing

Assessment 3

	Assessment Type: Individual assignment. Submit online via Canvas→Assignments→Assessment 3. Marks awarded for meeting requirements as closely as possible. **Clarifications/updates may be made via announcements/relevant discussion forums**.
	Due date: 23:59 PM on 12 September 2026. Please check Canvas→Assignments→Assessment3 for the most up-to-date information. A standard penalty of 10% per each working day past the deadline applies for up to 5 working days late unless a special consideration has been granted.
	Weighting: 40 marks

1. Overview

Assessment 3 will contribute 40% towards your final grade of this course and is focused on implementing an end-to-end cloud application using AWS cloud platform and technologies. You are free to use any programming language/framework and third-party API of your choice to develop your solution. You are provided with some sample solution architecture documents (see *Assessments -> Assessment 3* to download them) to help you understand the depth and required skills of the project, but **keep in mind that these samples just provide you references as to what a good project idea might look like and that the specification against which they were originally generated, might be different.** For instance, some of the previous iterations of the course required students to include a developer and user manual in a project report**. However, you are NOT required to provide any developer or user manuals as part of this assignment. Instead, you're expected to compile a properly scoped solution architecture document based on the guidelines provided in this specification.**

Before you develop your solution, discuss your proposal with your tutor. Aim to finalize your preliminary project idea by Week 5 or 6, so you can start early. You can refer the following example project ideas for inspiration. One example idea has been expanded to include potential use-cases that could be implemented and services that could be used.

- A contact tracing tool for health authorities and governments.
  Example use-cases that can potentially be implemented:
  - *Trace:* find out when you have been near other app users who have tested positive for coronavirus.
  - *Alert:* lets you know the level of coronavirus risk in your postcode district.
  - *Check-in:* get alerted if you have visited a venue where you may have come into contact with coronavirus, using a simple QR code scanner. No more form filling.
  - *Symptoms:* check if you have coronavirus symptoms and see if you need to order a test.
  - *Test:* helps you order a test if you need to.
  - *Isolate:* keep track of your self-isolation countdown and access relevant advice.

  AWS services that can potentially be used:
  - AWS RDS: To persist application state and data.

  - AWS Elastic Beanstalk: To achieve the ease of deployment and scaling.
  - AWS API Gateway: To expose application functionalities as APIs to user interfaces as well as third-party applications.
  - AWS Lambda: To implement application functionalities that get triggered on-demand in response to events from user interfaces.
  - AWS CloudFront: To efficiently deliver media content (e.g. images, etc) to the users.
  - AWS S3: To persist media content.
  - AWS IAM (No marks offered): To manage identities and access across the application.
  - AWS SES (No marks offered): To generate emails.

- A light-weight ride-sharing system that allows passengers to book rides, and drivers to accept bookings.
- A light-weight social media platform that allows users to share content within their social network.
- A light-weight crypto exchange platform that allows users to build and maintain portfolios of crypto assets, trade them and monitor trends.

A **recommended** timeline to help you finish your Assessment 3 on time can be found below:

Weeks	Milestones	Recommended Tasks
Week 4		AWS Service Investigation **References:** Lecture slides, AWS documentation
Week 5	Preliminary Project Idea Consultation (in Week 5 during the online student consultation sessions)	Preliminary Project Idea Formation **References:** Lecture sides, AWS documentation, tutorial notes, sample solution architecture documents from the past offerings
Week 6	System Architecture Consultation(in Week 6 during the online student consultation sessions)	Project Planning System Architecture Design **References:** Lecture sides, AWS documentation, tutorial notes, sample solution architecture documents from the past offerings
Week 7-12	Project Implementation Consultation (during online student consultation sessions)	Project Implementation/Testing Solution Architecture Document Writing **References:** AWS documentation, tutorial notes, sample solution architecture documents from the past offerings
Week 12	Project Implementation, Solution Architecture Document and Code Submission Due	Project Implementation Wrap-Up Solution Architecture Document Wrap-Up Submission **References:** AWS documentation, tutorial notes, sample solution architecture documents from the past offerings
Week 13	Project Demo Due	Project Demo

2. Assessment Criteria

This assessment will focus on:

1. Designing and developing a highly scalable application by applying the knowledge of distributed architecture and multiple cloud services.
2. Compiling an end-to-end document describing the selected *solution architecture* for your application.

3. Delivering a demo your product.

3. Learning Outcomes

This assessment is relevant to the following Learning Outcomes:

1. Develop and deploy a fully-fledged cloud application using popular cloud platforms.
2. Design and develop highly scalable cloud-based applications by creating and configuring virtual machines on the cloud and building private cloud.
3. Explain and identify the techniques of Bigdata analysis in cloud.
4. Compare, contrast, and evaluate the key trade-offs between multiple approaches to cloud system design, and identify appropriate design choices when solving real-world cloud computing problems.
5. Compile a comprehensive solution-architecture document.
6. Make recommendations on cloud computing solutions for an enterprise.

4. Assessment details

Criteria/Project requirements:

1. You must use the AWS services under the categories of **Compute, Containers, Storage, Networking and Content Delivery, Database,** and **Analytics** on the AWS Management Console for your application.
2. Each service/API must be 1) **fully implemented and automated** in your application and 2) **assessed as an appropriate selection for your application** by the examiner, in order to receive full marks.

   [Note:] Fully implemented and automated services/APIs mean the services/APIs should be automatically invoked by your client interface operations/code/other services other than CLI/AWS Console.
3. The following fully implemented AWS services are worth **6 marks per type**: Elastic Beanstalk, Lambda, API Gateway, ECS, and EMR.
4. The other fully implemented AWS services under the categories of Compute, Containers, Storage, Networking and Content Delivery, Database, and Analytics on the AWS Management Console for your application are worth **3 marks per type**.

   [Note:] The mark calculation for implemented AWS services **cannot be iterated**, e.g., if you fully implement an Elastic Beanstalk/ECS/EMR/Lambda service that automatically contains an EC2 service and an S3 service, you can only receive 6 marks in total.
5. You are encouraged to use third-party APIs external to AWS cloud services. Each type of fully implemented and automated API is worth **2 mark per type**. For example, if you fully implement and automate a Twitter API (e.g., Twitter Sign-in API) or a Facebook API (e.g., Facebook Feed API), 2 marks will be rewarded **per service type**.
6. You should have a well-designed and user-friendly **client-side interface** (e.g., a webpage/website or a mobile app). If you have any kind of data analysis you should interpret your result nicely using tabular and/or graphical format. You must deploy your application in AWS cloud.
7. Each student should submit the project with **a solution architecture** document and demo their project to be assessed (see *6. Submission, Demonstration, and Solution Architecture Document* for details).

Project Options

1. Option 1: Development of a cloud application using your own idea and strength (**Note:** You are **NOT** allowed to reuse your Assessment 2 application)
2. Option 2: Development of a cloud application using idea suggested by your tutor based on your interest and strength.

5. Referencing guidelines

What: All submitted contents must be your own. If you have used sources of information other than the contents directly under Canvas→Modules, you must give acknowledge the sources and give references using IEEE referencing style.

Where: Add a code comment near the work to be referenced and include the reference in the IEEE style.

How: To generate a valid IEEE style reference, please use the citethisforme tool if unfamiliar with this style. Add the detailed reference before any relevant code (within code comments).

6. Submission, Demonstration, and Solution Architecture Document

You need to submit a solution architecture document, and all the materials related to your project before the deadline outlined in Canvas->Assignments->Assessment 3 section. Any submission received after the aforementioned deadline will be considered as a late submission. **You will be penalized 10% of your total mark per working day for late submissions. You will not receive any marks if you do not submit your project materials and the solution architecture document.**

During submission you will need to provide the following content in a .zip file.

1. Your solution architecture document in a word or pdf document. You can make your own format or follow the given structure, if preferred.
   The solution architecture document should contain the following materials:
   a. **Links**: Live URL of your project, repository URL (eg. github/bitbucket/google drive/dropbox) of your source code (if any – but do not store your AWS credential information for security reasons), public dataset links of your project (if any).
   b. **Summary**: A short description on the objective/purpose of your project.
   c. **Introduction**: Introduce your project such as,
      i. **Motivation:** What are the key motivations?
      ii. **High-level view**: What does it do at a high-level (e.g., birds-eye view)?
      iii. **Beneficiaries:** Who are the key beneficiaries/target audiences? (e.g., types of users, organizations, etc.)
   d. **Related work**: Refer some related works similar to your application.
   e. **System Architecture**
      i. One or more comprehensively documented architectural diagram(s) that shows the function and communication between different cloud components used in your project. **[Note:]** The diagram(s) must be able to clearly show 1) the whole process how each client interface operation invokes the other components of the system, 2) the detailed interactions (e.g., invoking) among all the components, and 3) the functions of all the components. Some good examples are provided via *Assessments -> Assessment 3* section in Canvas.
      ii. A system description that explains the purpose of using each of those components.
      iii. Description of any datasets/data structures/APIs you used for your project (if any) [use figures if required]
   f. **References**: Important references/website links that you use to develop your application.

2. Put all the images you have used in your solution architecture document in a folder name doc_images.
3. Put all the source code of your project in a folder named code. If source code is greater than 5 MB then provide a google_drive/dropbox/github share link in a text file (name it **code.txt**). to download your source code (Note: DO NOT put any security credentials there).

4. Put runnable/deployable files (if any e.g. .war, .zip, .jar) in a folder named deploy.
5. Put all the data/SQL scripts (if any) in a folder named data. If data files are larger than 5MB then provide a link to download your data in a text file (name it data.txt)

You must demonstrate your project online (f2f option is also available) to your tutor in **Week 12**. The demo booking will be made available to students in **Week 12**. All demonstrations must be completed by **Week 12**. There will be a **penalty** if you fail to complete and demonstrate your work by **Week 12**.The demos will run for around **30 minutes** (including project introduction, app demo and Q&A) for each student. During demonstrations, you will also demo your app to the tutor as well as walk through your solutions architecture document. Keep everything ready and make your application live during your demo.

7. Academic integrity and plagiarism (standard warning)

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e., directly copied), summarized, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods,
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviors, including:

- Failure to properly document a source.
- Copyright material from the internet or databases.
- Collusion between students.

For further information on our policies and procedures, please refer to the University website.

8. Assessment declaration

When you submit work electronically, you agree to the assessment declaration.

9. Rubric/assessment criteria for marking

Criteria	Ratings			Pts
Project idea and project formulation with selection ofappropriate technologies	**2 Pts** **Project idea excellently formulated with selection of appropriate technologies.**	**1 Pts** **Project idea partially formulated with selection of most technologies being appropriate.**	**0 Pts** **Project idea not well formulated (with major issues) with selection of most technologies being inappropriate.**	2
Skill development in learning new tools and technologies for project completion	**3 Pts** **Demonstration of excellent knowledge in learning new tools and technologies for project completion.**	**1.5 Pts** **Demonstration of good knowledge (with minor issues) in learning new tools and technologies for project completion.**	**0 Pts** **Demonstration of poor knowledge (with major issues) in learning new tools and technologies for project completion.**	3
Appropriate utilization and full implementation of cloud tools/technologies/services in your project.	**35~0 Pts** **Appropriate utilization and full implementation of cloud services and/or APIs in your project according to Criteria 1-5.** **(In case students choose to integrate third-party APIs, although there is no limit to the number of such APIs they can integrate, only *two* would be graded.)**			25
Solution architecture document	**10~0 Pts** ***Summary (0.5 Pts)** ***Introduction (1 Pts)** ***Related work (1 Pts)** ***System architecture (5 Pts)**			10

	***System descriptions (1 Pts)** ***Description of any datasets/data structures/APIs (1 Pts)** ***References (0.5 Pts)** **Criteria:** **100% Pts -- A part is well written with substantial details and rich visualizations (if required).** **67% Pts – A part is written with some details and visualizations (if required).** **33% Pts -- A part is poorly written.** **0% Pts -- A part is missing.**	